

DAYTON WRIGHT BROTHERS, OH, USA

WMO: 724276

Lat: 39.594N

Lon: 84.226W

Elev: 962

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53859

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	9.5	-4.3	4.5	5.9	0.3	5.8	11.1	24.1	37.5	21.0	35.4	6.3	280	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	90.4	73.7	88.2	72.7	85.9	71.5	76.5	86.1	75.2	83.9	74.0	82.1	8.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	128.3	81.0	72.4	124.3	80.1	71.6	120.7	79.2	40.5	86.1	39.2	84.3	38.0	81.8	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.1	18.0	16.3	DB	-1.4	93.2	6.4	3.0	-6.0	95.4	-9.8	97.2	-13.4	98.8	-18.0	101.0
(5)				WB	-2.5	78.5	6.1	1.6	-6.9	79.7	-10.5	80.6	-13.9	81.5	-18.4	82.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	29.2	32.5	42.2	53.9	63.8	71.6	74.7	73.6	67.3	55.5	44.1	34.2
	DBStd	18.43	12.08	11.53	11.06	9.23	8.24	5.65	5.15	4.86	7.24	9.08	9.61	10.53
	HDD50	2290	652	500	296	64	5	0	0	0	0	46	228	499
	HDD65	5180	1111	910	709	345	126	13	1	3	59	320	628	955
	CDD50	3628	6	10	54	183	433	649	767	733	518	215	51	9
	CDD65	1040	0	0	2	13	89	212	303	270	126	24	1	0
	CDH74	8497	0	1	15	106	699	1665	2628	2169	1041	171	2	0
	CDH80	2650	0	0	1	8	158	504	915	693	334	37	0	0
Wind	WSAvg	7.7	9.1	8.9	8.9	9.1	7.4	6.6	6.0	5.9	6.3	7.3	8.2	8.5
Precipitation	PrecAvg	41.5	3.1	2.3	3.4	3.9	4.3	4.4	4.0	3.2	3.4	3.3	2.6	3.4
	PrecMax	59.8	8.0	4.7	7.7	10.3	7.0	7.8	6.3	6.4	8.1	5.5	6.2	5.9
	PrecMin	29.6	1.3	1.1	1.2	1.7	1.5	1.9	2.6	1.8	0.9	1.4	0.7	1.6
	PrecStd	6.3	1.7	1.2	1.4	1.9	1.8	1.6	1.1	1.2	2.0	1.4	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.9	67.8	75.8	81.2	87.6	91.0	93.9	92.3	91.7	84.6	72.0	63.4
		MCWB	57.5	58.8	60.7	64.3	71.4	73.1	76.4	73.6	71.5	69.4	60.1	57.7
	2%	DB	57.3	60.5	69.8	77.2	84.0	88.0	90.4	89.4	87.8	78.5	66.4	58.2
		MCWB	53.4	53.8	57.9	61.4	68.7	72.1	75.3	73.1	70.4	65.7	57.2	54.0
	5%	DB	53.0	55.3	64.3	73.3	81.4	85.4	88.1	86.5	83.9	74.6	63.3	54.2
		MCWB	49.3	49.8	54.6	59.3	67.2	70.9	73.7	72.0	69.4	63.4	56.0	50.4
	10%	DB	47.3	50.1	60.3	69.7	78.3	82.7	85.5	83.9	80.6	70.6	59.2	49.8
		MCWB	42.7	44.5	52.5	58.1	65.9	69.7	72.6	71.1	67.5	61.5	52.6	45.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.7	59.8	63.3	67.5	73.2	76.4	78.9	77.3	75.2	71.6	63.2	59.3
		MCDB	61.9	66.7	72.9	77.7	84.3	85.6	89.6	87.2	84.7	80.3	67.7	62.2
	2%	WB	54.7	55.4	59.6	64.4	71.0	74.5	77.2	75.7	73.2	68.8	59.9	54.8
		MCDB	57.1	58.9	66.8	71.8	80.8	83.9	87.5	84.8	82.5	76.0	64.9	57.4
	5%	WB	49.8	50.5	57.0	62.0	69.2	73.2	75.8	74.5	71.5	65.7	56.9	50.9
		MCDB	52.4	55.3	63.1	69.1	78.1	82.3	84.8	82.6	80.6	72.2	61.9	54.0
	10%	WB	43.1	44.3	53.4	59.8	67.5	71.7	74.5	73.1	69.8	62.6	53.5	45.9
		MCDB	47.0	49.4	59.3	67.2	75.1	79.7	82.4	80.5	76.8	69.1	58.5	49.1
Mean Daily Temperature Range	5% DB	MDBR	14.1	15.5	17.1	19.8	18.7	18.0	17.6	18.1	19.5	18.4	16.4	13.0
		MCDBR	17.9	21.4	22.4	23.9	21.4	20.8	19.3	20.5	22.9	22.2	20.5	17.5
		MCWBR	16.0	17.3	14.0	13.6	10.0	9.2	7.6	8.2	9.6	11.8	14.1	15.5
	5% WB	MCDBR	17.3	20.5	19.7	19.7	18.3	17.6	17.4	16.8	18.3	18.5	17.3	16.8
		MCWBR	16.5	17.7	13.6	13.1	9.3	8.9	7.6	7.7	8.7	10.8	13.8	16.1
Clear-Sky Solar Irradiance	taub	0.300	0.323	0.346	0.389	0.434	0.444	0.447	0.442	0.392	0.352	0.322	0.300	
	taud	2.462	2.364	2.378	2.268	2.184	2.202	2.202	2.198	2.315	2.428	2.472	2.476	
	Ebn at Noon	277	284	288	281	269	266	264	262	269	270	266	268	
	Edh at Noon	25	32	35	42	46	45	45	44	37	30	25	23	
All-Sky Solar Radiation	RadAvg	550	815	1137	1535	1742	1928	1875	1734	1421	1008	681	465	
	RadStd	48	97	98	124	145	122	111	89	135	95	80	46	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212	+126

Nomenclature: See separate page